# 动理学方程的高效特征线型渐近保持方法

蔡逸[1]，洪逸扬[1]，熊涛[2,*]

（1，厦门大学数学科学学院，福建 厦门 361005；
2，中国科学技术大学数学科学学院，安徽 合肥 230026）

**摘　要**：动理学方程是描述微观粒子运动规律的重要数学模型，其多尺度和高维度等特性为数值格式的设计带来显著挑战．渐近保持格式是一类重要的多尺度计算框架，旨在整个计算域内使用统一的数值求解器，自动捕捉方程的渐近极限．近年来，我们以特征线回溯思想为基础，发展了一类全尺度一致无条件稳定的渐近保持算法框架，它具备无条件稳定、计算复杂度低、高度可并行和良好可扩展等优点，特别适合高维跨尺度输运问题的长时间数值模拟，将在国防工程、天体物理、半导体和生物化学等领域有广阔的应用前景．

**关键词**：动理学方程；并行计算；特征线回溯法；渐近保持格式

**中图分类号**：O241.8　　　　**文献标识码**：A

## 0　引言

动理学方程是描述微观粒子运动规律的重要数学模型，在辐射输运[1]、等离子体物理[2]、半导体[3]和生物化学[4]等领域具有广泛应用．粒子在自由输运时不断发生碰撞，其演化过程在统计意义下可由微分方程描述：

$$\frac{\partial f}{\partial t}+T(f)=C(f).$$

其中，$f=f(t,x,v)$ 为粒子分布函数，$t\geq 0$ 为时间变量，$x\in\mathbb{X}$ 为位置变量，$v\in\mathbb{V}$ 为速度变量，$\mathbb{X}\times\mathbb{V}$ 为相空间，$T$ 为输运算子，$C$ 为碰撞算子．

动理学方程具有双曲–抛物混合特性．传统数值离散多采用直线型方法(method of lines)：先离散相空间，再离散时间．时间离散常采用隐式方法、预估矫正法或特征线法，以缓解刚性项带来的时间步长约束．相空间离散可分为确定性方法与随机性方法．在确定性方法中，矩方法(moment method)和离散速度法(discrete velocity method)均是先对速度变量 $v$ 采用全局近似，再对位置变量 $x$ 采用分片近似．矩方法[5,6]通过对速度变量进行矩展开，利用正则化思想构造矩封闭条件[7–11]，将高维动理学模型约简成低维宏观模型．然而，宏观模型的输运结构更加复杂，在输运占优问题中易出现强波动性[12,13]．离散速度法[14]在有限个离散速度点上直接求解动理学方程，可严格保持输运结构．但是，由于信息传播方

**Commented [1]:** 正文和公式中如有向量,矩阵,矢量，张量这类变量请采用黑体+斜体表示。请作者注重该项检查，避免后续增加返修次数。

**Commented [逸蔡2R1]:** 算子符号已统一采用大写字母表示

**收稿日期**：2025#-##11-##21；　**修回日期**：2025#-00-00

**基金项目**：国家重点研发计划重点专项项目（2022YFA1004500）、国家自然科学基金项目（12571443）、福建省自然科学基金（2023J02003）

**作者简介**：蔡逸，博士研究生，研究方向为动理学方程渐近保持间断 Galerkin 有限元方法. E-mail: yicaim@stu.xmu.edu.cn
~~洪逸扬，博士研究生，研究方向为辐射输运方程渐近保持蒙特卡罗方法. E-mail: hongyy23@stu.xmu.edu.cn~~

***通信作者**：熊涛，博士，教授，博士生导师，研究方向为动理学方程的高效算法．　E-mail: taoxiong@ustc. edu. cn

向受限于预设速度集，当速度点数不足时易出现射线效应[15,16]. 随机性方法的代表为蒙特卡罗法[17] (Monte-Carlo method). 该方法~~根据方程的积分形式~~将动理学方程等价表述为粒子随机运动过程，利~~采~~用随机采样构造数值解，具有无网格性与高度并行性，但在强碰撞区域计算效率会显著下降.

动理学方程具有多尺度特征. 为分析其长时间演化行为，可通过量纲分析引入无量纲参数 Knudsen 数 $\varepsilon$，并考虑下列多尺度模型:

$$\varepsilon\frac{\partial f}{\partial t}+T(f)=\frac{\varepsilon}{\lambda(\varepsilon)}C(f).$$

当$\lambda(\varepsilon)\sim\varepsilon$时，输运效应与碰撞效应处于相同量级，当$t/\varepsilon\to+\infty$时，方程收敛至稳态极限. 当$\lambda(\varepsilon)\sim\varepsilon^2$时，碰撞效应显著增强但仍不足以完全抹平输运效应，方程呈现双曲-抛物混合特性: 当$\varepsilon\gg 0$时，方程是双曲占优的; 当$\varepsilon\approx 0$时，方程是抛物占优的; 当$t\to+\infty$时，方程仍然收敛至稳态极限. 为构造能够同时适用于不同区域的数值方法，现有研究主要沿两个方向发展: 区域分解(domain decomposition)与渐近保持(asymptotic-preserving). 区域分解方法[18–24]将计算域划分为不同的物理子区域，在各区域分别采用适合的数值方法求解，并通过界面条件实现整体匹配. 渐近保持方法[25]旨在整个计算域内采用统一的数值方法，在离散意义下自动捕捉方程的渐近极限. 典型的渐近保持格式有: 1) 松弛格式[26,27]; 2) 宏微观分解[28–34]; 3) 奇偶分解[35,36]; 4) 罚函数法[37,38]; 5) 指数积分法[39–43]等等.

近年来，我们以线性动理学方程为出发点，围绕特征线回溯法开展了一系列研究，发展了一类全尺度无条件稳定、高度可并行且易扩展的渐近保持算法，并进行了相应的理论稳定性分析. 同时，我们进一步将该算法推广应用于非线性辐射输运方程，取得了良好的数值效果.

## 1　基于特征线回溯法的渐近保持格式

对于具有抛物极限的多尺度动理学模型，其输运项和碰撞项在$\varepsilon\approx 0$时均具有强刚性，这为时间离散带来了显著的数值挑战.一方面，根据 CFL 条件，显式格式受到非常严苛的时间步长约束$\tau\sim\varepsilon h$，其中$\tau$为时间步长，$h$为空间尺寸; 另一方面，隐式格式虽然无条件稳定，但在高维情形下需求解大规模强耦合线性系统，计算成本极其高昂. 为平衡稳定性和计算成本，Peng 等[32–34]构造了一类显隐格式，通过隐式处理扩散相关项，实现抛物占优区域($\varepsilon\ll h$)无条件稳定. 但是，在过渡区域($\varepsilon\sim h$)仍然会受限于抛物型时间步长约束($\tau\sim h^2$)对跨区域问题而言仍存在明显局限. 注意到这一瓶颈的根源在于输运项的刚性，我们发展了一类基于特征线回溯法的渐近保持格式[44,45]，其具备如下优势:

1) 全尺度一致无条件稳定性且计算复杂度低. 与显式格式相比, 该方法在理论上不受 CFL 步长约束; 与隐式格式相比, 仅需求解少量小规模宏观线性系统, 计算代价显著降低; 与显隐格式相比, 在过渡区域仍可保持无条件稳定, 从而显著提升跨区域问题的计算效率;
2) 高度并行性. 在传统隐式迎风格式中, 由于目标时间层单元在位置变量上具有强耦合性, 导致算法关于位置变量的并行效率低下. 而特征线回溯法可完全解耦目标时间层单元, 使算法在整个相空间上具有近似线性的加速比;
3) 良好的可扩展性与兼容性. 该方法对具有刚性输运项的动理学模型均具有普适性. 此外, 我们已经将其应用至蒙特卡罗法[46, 47]和矩方法[48]框架, 显著提升了相应方法在长时间模拟时的稳定性和计算效率.

### 1.1 模型重构

下面我们以一维方程为例介绍基于特征线法的模型重构过程. 设 $E(v)$ 是全局平衡态函数, 满足归一化条件和对称性条件:

$$\int_{\mathbb{V}} E(v)\mathrm{d}v = 1, \quad \int_{\mathbb{V}} vE(v)\mathrm{d}v = 0. \tag{1}$$

**Commented [3]:** 全文公式请统一从（1）开始编号

**Commented [逸蔡4R3]:** 已重新编号

设 $f_{eq}(t,x,v) = \rho(t,x)E(v)$, $\rho(t,x) = \int_{\mathbb{V}} f(t,x,v)\mathrm{d}v$, 考虑一维线性输运方程:

$$\varepsilon \frac{\partial f}{\partial t}(t,x,v) + v\frac{\partial f}{\partial x}(t,x,v) = \frac{\varepsilon\sigma}{\lambda(\varepsilon)}(f_{eq}(t,x,v) - f(t,x,v)). \tag{2}$$

设 $j(t,x) = \int_{\mathbb{V}} vf(t,x,v)\mathrm{d}v$, 系统满足局部质量守恒律:

$$\varepsilon \frac{\partial \rho}{\partial t}(t,x) + \frac{\partial j}{\partial x}(t,x) = 0. \tag{3}$$

设 $f_{\perp}(t,x,v) = f(t,x,v) - f_{eq}(t,x,v)$, 由 $f_{eq}$ 的定义和 $E$ 的归一化条件得

$$\int_{\mathbb{V}} f_{\perp}(t,x,v)\mathrm{d}v = 0. \tag{4}$$

引入参数 $\tau$, 由特征线法得

$$f(t,x,v) = e^{-\sigma\tau/\lambda} f(t-\tau, x - \frac{v}{\varepsilon}\tau, v) + \int_0^{\tau} \frac{\sigma}{\lambda} e^{-\sigma(\tau-s)/\lambda} f_{eq}(t-\tau+s, x - \frac{v}{\varepsilon}(\tau - s), v)\mathrm{d}s. \tag{5}$$

引入参数 $\theta = \theta(\sigma,\lambda) \in [0,1]$ 满足: 1) 当 $\sigma \to 0$ 时, $\theta \to 1$; 2) 当 $\lambda \sim \varepsilon$ 时, $\theta \approx 1$; 3) 当 $\lambda \sim \varepsilon^2$ 或 $\sigma \to +\infty$ 时, $\theta \approx 0$. 由(5)可将 $f_{\perp}(t,x,v)$ 分解为凸组合

$$f_{\perp}(t,x,v) = \theta f_{\perp,H}(t,x,v) + (1-\theta) f_{\perp,P}(t,x,v). \tag{6}$$

一方面, $f_{\perp,H}(t,x,v)$ 用于捕捉自由输运极限和稳态极限, 写作

$$f_{\perp,H}(t,x,v)=e^{-\sigma\tau/\lambda}(f_{\perp}+f_{eq})(t-\tau,x-\frac{v}{\varepsilon}\tau,v)-e^{-\sigma\tau/\lambda}f_{eq}(t,x,v)+r_H(t,x,v). \tag{7}$$

定义

$$\varphi_k(z)=\begin{cases}1, & k=0;\\ \int_0^1\frac{\omega^{k-1}}{(k-1)!}e^{\omega z}d\omega, & k\ge 1.\end{cases} \tag{8}$$

其满足

$$\varphi_k(0)=\frac{1}{k!},\quad \forall k\ge 0, \tag{9}$$

且

$$\varphi_k(z)=\frac{e^z\varphi_{k-1}(0)-\varphi_{k-1}(z)}{z},\quad \forall k\ge 1. \tag{10}$$

记$\phi_k(z)=z^k\varphi_k(z)$，$D^k=(\frac{\lambda}{\sigma})^k(\frac{\partial}{\partial t}+\frac{v}{\varepsilon}\frac{\partial}{\partial x})^k$，则残差项$r_H(t,x,v)$可写作

$$r_H(t,x,v)=-e^{-\sigma\tau/\lambda}\sum_{k\ge 1}\phi_{k+1}(-\frac{\sigma\tau}{\lambda})D^k f_{eq}(t,x,v). \tag{11}$$

另一方面，$f_{\perp,P}(t,x,v)$由分部积分法得到，用于捕捉抛物极限，写作

$$f_{\perp,P}(t,x,v)=e^{-\sigma\tau/\lambda}f_{\perp}(t-\tau,x-\frac{v}{\varepsilon}\tau,v)-\frac{\lambda}{\sigma}(1-e^{-\sigma\tau/\lambda})(\frac{\partial f_{eq}}{\partial t}+\frac{v}{\varepsilon}\frac{\partial f_{eq}}{\partial x})(t,x,v)-r_P(t,x,v), \tag{12}$$

其中，残差项$r_P(t,x,v)$可写作

$$r_P(t,x,v)=-\frac{\lambda}{\sigma}e^{-\sigma\tau/\lambda}\sum_{k\ge 1}\phi_{k+1}(-\frac{\sigma\tau}{\lambda})D^{k+1}f_{eq}(t,x,v). \tag{13}$$

将上述特征线解代入(3)，联立(2)可得

$$\begin{cases}\frac{\partial f}{\partial t}(t,x,v)+\frac{v}{\varepsilon}\frac{\partial f}{\partial x}(t,x,v)=\frac{\sigma}{\lambda}(f_{eq}(t,x,v)-f(t,x,v)),\\ \frac{\partial\rho}{\partial t}(t,x)+\frac{1}{\varepsilon}e^{-\sigma\tau/\lambda}\int_{\mathbb{V}}v\frac{\partial(f_{\perp}+\theta f_{eq})}{\partial x}(t-\tau,x-\frac{v}{\varepsilon}\tau,v)\mathrm{d}v=\frac{\lambda}{\varepsilon^2\sigma}(1-\theta)(1-e^{-\sigma\tau/\lambda})\Theta\frac{\partial^2\rho}{\partial x^2}(t,x)+R(t,x).\end{cases} \tag{14}$$

其中，$\Theta=\int_{\mathbb{V}}v^2E(v)\mathrm{d}v$为扩散系数，$R(t,x)$为残差项，写作

$$R(t,x)=\int_{\mathbb{V}}v(\theta r_H(t,x,v)-(1-\theta)r_P(t,x,v))dv. \tag{15}$$

当$\lambda(\varepsilon)=\varepsilon$时，有$\theta=1$，

$$f(t,x,v)\xrightarrow{t/\varepsilon\to+\infty}f^{\infty}(v)=\rho^{\infty}(x)E(v). \tag{16}$$

此时，$\rho^{\infty}$是由初值条件和边界条件确定的常数. 当$\lambda(\varepsilon)=\lambda_0\varepsilon^2$时，有$\theta\approx 0$，

$$f(t,x,v)\xrightarrow{\varepsilon\to 0}f^{\lambda_0}(t,x,v)=\rho^{\lambda_0}(t,x)E(v), \tag{17}$$

其中，$\rho^{\lambda_0}$ 满足抛物方程

$$\frac{\partial \rho^{\lambda_0}}{\partial t}(t,x)=\frac{\lambda_0}{\sigma}\Theta\frac{\partial^2\rho^{\lambda_0}}{\partial x^2}(t,x). \tag{18}$$

再令 $t\to+\infty$ 或 $\lambda_0\to+\infty$，可得 $\rho^{\lambda_0}(t,x)\to\rho^{\infty}(x)$. 对于更复杂的碰撞算子，一种可行的策略是采用线性加罚对模型进行重构. 以各向异性散射为例，其碰撞算子可写为

$$C(f)=\int_{\mathbb{V}}\sigma(v,v')(f(t,x,v')-f(t,x,v))\mathrm{d}v',\quad \int_{\mathbb{V}}\sigma(v,v')\mathrm{d}v'=1. \tag{19}$$

可将其分解为

$$C(f)=P(f)+Q(f),\quad P(f)=C(f)-Q(f), \tag{20}$$

其中，$P(f)=\sigma_0(f_{eq}(t,x,v)-f(t,x,v))$ 为罚项，$Q(f)$ 为源项. 应用前文所述的模型重构过程，便能构造出渐近保持的宏观-微观输运模型.

**1.2　数值格式**

下面我们以一阶有限体积格式为例描述数值格式的构造过程. 设 $\tau_n$ 为第 $n$ 个时间步的步长，$\tau=\max_n\tau_n$，$x_i$ 是单元 $I_i=[x_{i-1/2},x_{i+1/2}]$ 的中点，$h_i=x_{i+1/2}-x_{i-1/2}$，$h=\max_i h_i$，$(v_k,w_k)$ 是第 $k$ 个离散速度点与其对应的积分权重，$\rho_i^n$ 是变量 $\rho$ 在 $t_n$ 时刻单元 $I_i$ 上的均值，$f_{i,k}^n$ 是变量 $f$ 在 $t_n$ 时刻单元 $I_i$ 上第 $k$ 个离散速度分量的均值，$T_k^n x=x-\frac{v_k}{\varepsilon}\tau_n$ 是位置点 $x$ 沿第 $k$ 个离散速度回溯至 $t_n$ 时刻的上游位置点.~~-~~数值格式的构造分为两个步骤: 宏观方程的离散和微观方程的离散. 整体算法流程可见图 1.~~:~~

1) 宏观方程的离散. 时间上采用向后欧拉时间离散，可得:

$$\frac{\rho_i^{n+1/2}-\rho_i^n}{\tau_n}+\frac{1}{\varepsilon}e^{-\sigma\tau_n/\lambda}\sum_k w_k v_k S_{h,k}^n(f_{\perp,i,k}^n+\theta f_{eq,i,k}^n)=\frac{\lambda}{\varepsilon^2\sigma}(1-\theta)(1-e^{-\sigma\tau_n/\lambda})\Theta D_h\rho_i^{n+1/2}. \tag{21}$$

这里可以看到，我们对扩散项 $D_h\rho$ 采用隐式时间离散，因而在抛物极限处能够克服抛物型时间步长约束 $\tau\sim h^2$. 在宏观数值格式的构造过程中，离散输运算子 $S_{h,k}^n$ 的构造是关键，下面我们介绍两种典型思想:

a) 短程回溯. 该思想源自统一气体动理学格式[39,43]，通过沿局部特征线回溯重构界面通量.记 $g=f_\perp+\theta f_{eq}$，定义

$$S_{h,k}^n g_{i,k}^n = \begin{cases} \frac{1}{h_i}(g_{i+1/2,k}^{n,l} - g_{i-1/2,k}^{n,l}), & v_k > 0; \\ \frac{1}{h_i}(g_{i+1/2,k}^{n,r} - g_{i-1/2,k}^{n,r}), & v_k < 0. \end{cases} \tag{22}$$

其中，$g_{i+1/2,k}^{n,l}$是通量左侧重构值，$g_{i+1/2,k}^{n,r}$是通量右侧重构值. 对于一阶格式而言，可取$g_{i+1/2,k}^{n,l} = g_{i,k}^n$和$g_{i+1/2,k}^{n,r} = g_{i+1,k}^n$. 由于短程回溯时的特征线截断造成数值依赖域缩小，导致该方法在过渡区域的稳定性条件为$\tau \sim \varepsilon h$.

b) 长程回溯[44,45]. 该思想源自半拉格朗日 (Semi-Lagrangian) 格式[49,50]，通过严格追踪特征线至上游时间层，利用上游单元重构追踪点处的信息. 因为数值依赖域严格包含微分方程依赖域，所以该方法的稳定性不受时间步长约束. 假设$T_k^n x \in I_m$，$g_{*,k}^n(x)$是基于$\{g_{i,k}^n\}$信息的重构多项式，满足对于所有$\ell \in \{0,1\}$，

$$\begin{cases} \frac{1}{h_{m-\ell}} \int_{I_{m-\ell}} g_{*,k}^n(x) \mathrm{d}x = g_{m-\ell,k}^n, & v_k > 0; \\ \frac{1}{h_{m+\ell}} \int_{I_{m+\ell}} g_{*,k}^n(x) \mathrm{d}x = g_{m+\ell,k}^n, & v_k < 0. \end{cases} \tag{23}$$

由此可定义如下离散输运算子:

$$S_{h,k}^n g_{i,k}^n = \begin{cases} \frac{1}{h_i}(g_{*,k}^n(T_k^n(x_{i+1/2}^-)) - g_{*,k}^n(T_k^n(x_{i-1/2}^-))), & v_k > 0; \\ \frac{1}{h_i}(g_{*,k}^n(T_k^n(x_{i+1/2}^+)) - g_{*,k}^n(T_k^n(x_{i-1/2}^+))), & v_k < 0. \end{cases} \tag{24}$$

2) 微观方程的离散. 对于无条件稳定的微观数值格式的构造过程，我们介绍两种研究思路:

a) 全隐格式. 时间上采用向后欧拉时间离散，可得:

$$\frac{f_{i,k}^{n+1} - f_{i,k}^n}{\tau_n} + \frac{v_k}{\varepsilon} B_{h,k} f_{i,k}^{n+1} = \frac{\sigma}{\lambda}(f_{eq,i,k}^{n+1} - f_{i,k}^{n+1}). \tag{25}$$

其中，$B_{h,k}$是迎风离散算子，其一阶形式如下:

$$B_{h,k} f_{i,k}^n = \begin{cases} \frac{1}{h_i}(f_{i,k}^n - f_{i-1,k}^n), & v_k > 0; \\ \frac{1}{h_i}(f_{i+1,k}^n - f_{i,k}^n), & v_k < 0. \end{cases} \tag{26}$$

对于输运方程离散后得到的线性代数系统，常采用源迭代(source iteration)方法求解. 该方法的核心思想是预估$f_{eq}^{n+1}$，解耦各离散速度点对应的离散方程，从而实现逐速度的并行求解. 为保证迭代法的收敛性，为确保源迭代的收敛性，

需要构造合适的预条件子, 最典型的策略是合成加速型方法[51–54](synthetic acceleration). 为进一步提高迭代效率, 一方面可利用历史信息构造更优的搜索方向, 例如 Krylov 子空间方法[55,56]; 另一方面也可借助多重网格方法[57,58]在不同网格层次间传递误差以加速收敛. 此外, 若能获得高质量的初始迭代解, 也可显著减少迭代次数. 在我们的算法框架中, 宏观方程提供了高质量的数值局部平衡态 $f_{eq}^{n+1/2}$, 可视为特殊的初始迭代解, 仅需单步更新即可将残差降至目标精度. 关于输运方程迭代法的系统介绍, 可参考综述文章[59].

b) 半拉格朗日格式. 时间上采用向后欧拉格式,

$$f_{i,k}^{n+1}=\frac{\lambda}{\lambda+\sigma\tau_n}S_{i,k}^{n}f_{i,k}^{n}+\frac{\sigma\tau_n}{\lambda+\sigma\tau_n}f_{eq,i,k}^{n+1}. \tag{27}$$

记 $f_{*,k}^{n}(x)$ 是基于 $\{f_{i,k}^{n}\}$ 信息的重构多项式, 由此可定义离散输运算子:

$$S_{h,k}^{n}f_{i,k}^{n}=f_{*,k}^{n}(T_k^n(x_i)). \tag{28}$$

当 $f_{eq}^{n+1}$ 给定时, 微观量的更新无需求解线性方程组, 且 $t_{n+1}$ 时刻各空间单元之间不存在依赖关系, 从而能够实现逐单元、逐速度的完全并行求解, 特别适合在大规模并行集群上执行[60,61]. 若令 $f_{eq}^{n+1}=f_{eq}^{n+1/2}$, 仅需单步更新即可得到目标精度的微观数值解. 关于高精度半拉格朗日格式的构造, 可参考文献[62–64].

c) 蒙特卡罗格式. 假定数值解有如下表示:

$$f(t,x,v)=\sum_{p}\omega_p(t)\delta(x-x_p(t))\delta(v-v_p(t)), \tag{29}$$

其中, $x_p(t)$, $v_p(t)$ 和 $\omega_p(t)$ 分别是粒子 $p$ 在 $t$ 时刻的位置, 速度和权重, 它们满足以下关系:

$$\frac{dx_p}{dt}=\frac{v_p}{\varepsilon},\quad \frac{dv_p}{dt}=0,\quad \frac{d\omega_p}{dt}=-\frac{\sigma}{\lambda}\omega_p. \tag{30}$$

粒子的初始位置, 速度和时间通过随机抽样确定, 而其初始权重则依据给定的初值条件、边值条件及预估值 $f_{eq}^{n+1}$ 进行分配. 与需要根据指数分布解析处理粒子散射的传统蒙特卡罗方法相比, 本方法无需显式模拟每次散射, 而是将散射项拆分为阻尼项和源项. 这一改进在方差略有增加的情况下, 显著改善了计算效率, 达成了良好的性能权衡.

需要指出的是, 对于宏观方程(21)中的对流项计算, 我们可采用蒙特卡罗方法进行求解. 记 $g=f_{\perp}+\theta f_{eq}$, 可定义离散输运算子如下:

$$\overline{S}_h^n g_i^n = \frac{1}{\tau_n h_i}\left(\sum_{p\in A_{i+1/2}^n} sign(v_p)\omega_p - \sum_{p\in A_{i-1/2}^n} sign(v_p)\omega_p\right), \tag{31}$$

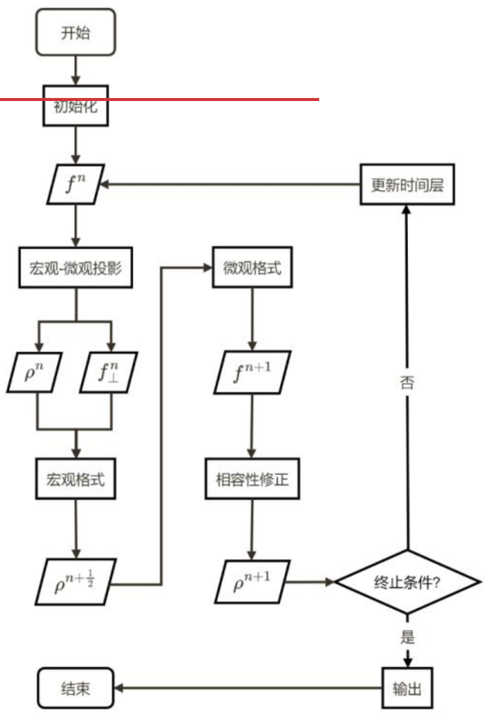
其中，$A_{i+1/2}^n$ 表示在时间区间 $[t_n, t_{n+1}]$ 内穿过 $x_{i+1/2}$ 的所有粒子，包括来自上一层的源粒子，边界条件粒子以及由 $f_{eq}$ 在 $t_n$ 时刻放出的粒子. 从而，宏观离散格式可改写为:

$$\frac{\rho_i^{n+1/2} - \rho_i^n}{\tau_n} + \frac{1}{\varepsilon}\overline{S}_h^n g_i^n = \frac{\lambda}{\varepsilon^2\sigma}(1-\theta)(1-e^{-\sigma\tau_n/\lambda})\Theta D_h \rho_i^{n+1/2}. \tag{32}$$

关于蒙特卡罗方法的实现细节，可参考文献[65,66].

3) 宏观-微观相容性. 若直接令 $\rho^{n+1} = \rho^{n+1/2}$，将会造成数值模拟的不稳定，其根源在于宏观量与微观量之间缺乏相容性，即，$\rho^{n+1/2} \neq \sum_k w_k f_k^{n+1}$. 为此，我们需要利用微观量对宏观量进行相容重构:

$$\rho_i^{n+1} = \sum_k w_k f_{i,k}^{n+1}. \tag{33}$$



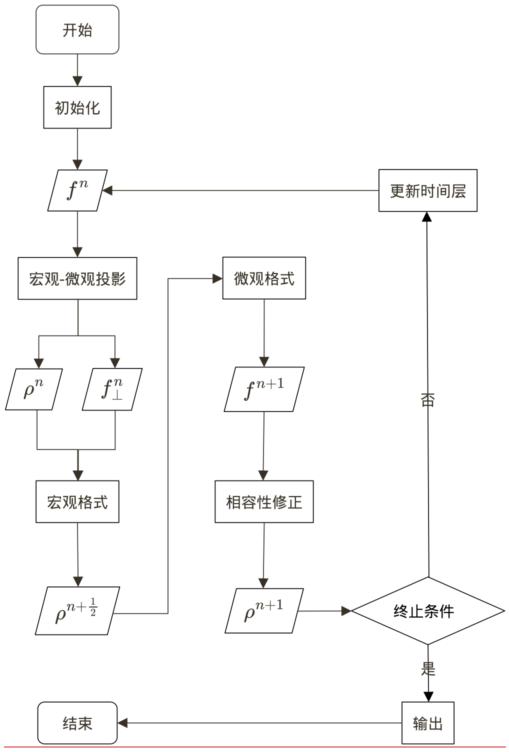



图 1 算法流程图.

Fig. 1 Algorithm flowchart.

**Commented [5]:** 缺少对图 1 的引用引用

**Commented [逸蔡6R5]:** 已在 1.2 节第一段话末尾添加对图 1 的引用

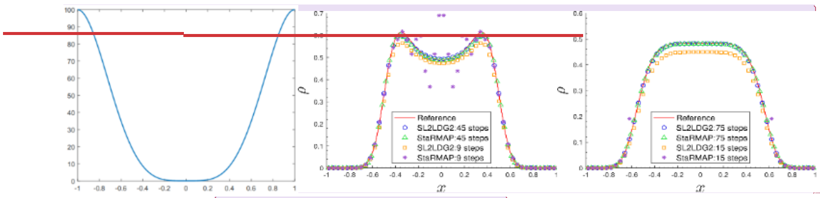
## 2 在辐射输运问题中的应用

### 2.1 线性输运方程

下面我们给出线性输运方程的部分三维数值算例. 间断有限元方法是求解双曲型微分方程的一类高精度数值方法, 其在并行计算、处理复杂几何和自适应网格细化(hp-adaptivity)等方面具有天然优势. 近年来, 针对动理学方程的渐近保持半拉格朗日格式得到广泛研究, 详见相关文献[67–70]等，和本文不同的是其考虑的极限不包含对流速度的刚性; 关于半拉格朗日间断有限元格式的具体构造, 可参考文献[71–73]. 记具有 $p$ 阶时间精度和 $q$ 阶空间精度的半拉格朗日间断有限元方法为SL$p$LDG$q$ [45], 并对比基于交错网格的二阶显式球谐函数法StaRMAP [74]. 本节展示的数值结果均来源于文献[45].

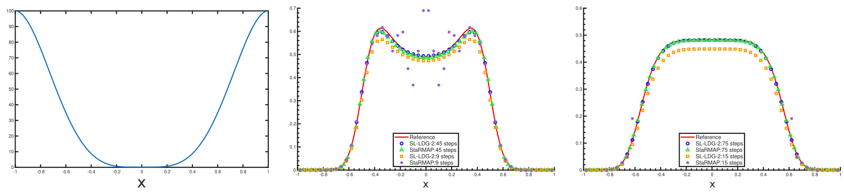
1) 三维变散射系数问题. 由图 2 所示, 对于具有剧烈空间变化的散射系数分布, 特征线方法利用沿特征方向追踪信息的机制, 在大时间步长约束的条件下仍能保持稳定可靠的计算结果; 相比之下, 传统显式格式在相同时间步长条件下

呈现数值不稳定性，难以得到有效的数值结果.



~~图 2  三维变散射问题. 左：散射系数分布图. 中，右：特征线法与球谐函数法的对比图.~~

~~Fig. 2 Three-dimensional varying scattering problem. Left: Scattering distribution. Middle, Right: Comparison of characteristic and spherical harmonic solutions.~~

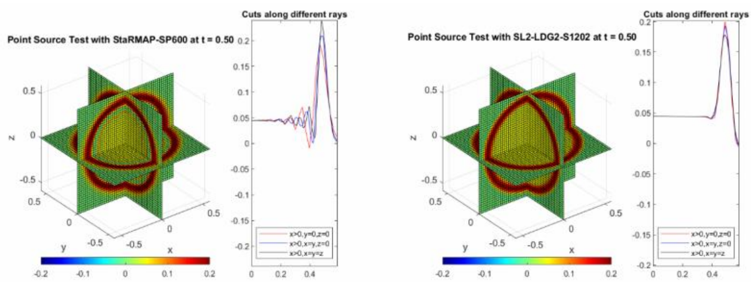


(a) (b) (c)

图 2  三维变散射问题. (a) 散射系数分布图; (b)(c)特征线法与球谐函数法的对比图.

Fig. 2 Three-dimensional varying scattering problem. (a) Scattering distribution; (b)(c) Comparison of characteristic and spherical harmonic solutions.

2) 三维点源问题. 由图 3 所示，当分布函数剧烈变化时，球谐函数法虽然能够保持解的旋转不变性，但是容易产生非物理数值振荡；而离散速度法由于严格保持输运结构，不容易产生波动. 由表 1 可见，本文所提方法的并行效率高，其加速比随计算核心数增加依然接近线性增长，验证了前文的理论分析结果.




~~图 3  三维点源问题($\sigma=1$). 左：球谐函数法；右：离散速度法.~~

~~Fig. 3 Three-dimensional point source problem ($\sigma=1$). Left: spherical harmonic method; Right: discrete velocity method.~~

**Commented [7]:** 优先提供矢量图（如 EPS/EMF/AI/CDR 格式），确保线条、文字可编辑。为确保正常显示，稿件正文中可保持当前图片格式不变。矢量图请单独整理到另外一份 word 文件中。形成单独的图文件。若无矢量图，需提交高分辨率位图（≥300dpi）。

**Commented [逸蔡8R7]:** 已在图文件提供 eps 文件

**Commented [9]:** 分图请改用（a）,(b),(c)等标注。**全文请照此修改，前后文不再一一特殊标注。**

**Commented [逸蔡10R9]:** 已修改分图标注

**Commented [11]:** 分图请改用（a）,(b),(c)等标注。**全文请照此修改，前后文不再一一特殊标注。**

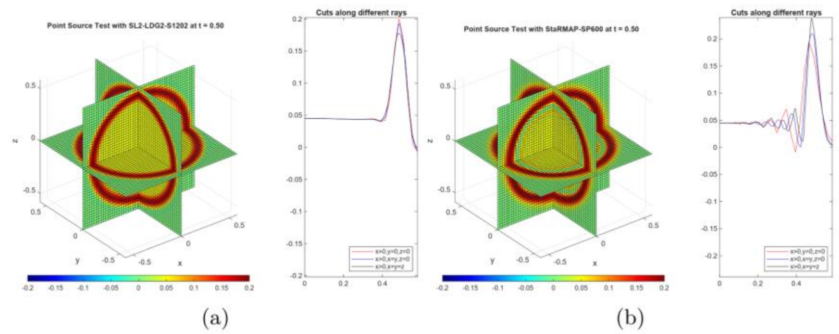



图 3 三维点源问题( $\sigma=1$ ). (a) 球谐函数法; (b)离散速度法.

Fig. 3 Three-dimensional point source problem ( $\sigma=1$ ). (a) Spherical harmonic method; (b) Discrete velocity method.

Field Code Changed

Field Code Changed

表 1 三维点源问题的 OpenMP 并行效率测试.

Table 1 Parallel efficiency test for the three-dimensional point source problem with OpenMP.

| 线程数 | 运行时间(秒) | 加速比 | 并行效率 |
|---|---|---|---|
| 1 | 13552.66 | - | - |
| 2 | 6835.17 | 1.98 | 99.14% |
| 4 | 3512.93 | 3.86 | 96.45% |
| 8 | 1641.28 | 8.26 | 103.22% |
| 16 | 1047.95 | 12.93 | 80.83% |

### 2.2 非线性辐射输运方程

对于非线性辐射输运方程，基于粒子的蒙特卡罗方法是主流的数值模拟工具之一. 在源项的离散上，显式蒙特卡罗方法受到极为严格的时间步长限制. 目前，隐式蒙特卡罗(IMC, Implicit Monte Carlo)[66]是应用最为广泛的算法. 该方法将光子的吸收与再发射过程近似为“伪散射”，虽显著增强了计算的稳定性，但在光学厚介质区域会引入高昂的计算开销. 为提高计算效率并规避伪散射的引入，我们将基于特征线近似的宏观模型与传统蒙特卡罗方法相结合，构造了新型蒙特卡罗方法[46,47](EMC)，并对比具有相同参数的 IMC 方法.

1) Infinite medium 问题(方差测试). 本算例求解的是灰体辐射输运方程，数值结果来源于文献[47]. 我们模拟一个处于扩散尺度的无限平板问题，真实解应该保持恒定温度. 定义方差与计算时间的性能指标(FOM, Figure of Merit)为

$$FOM=\frac{1}{VAR\times RT},$$

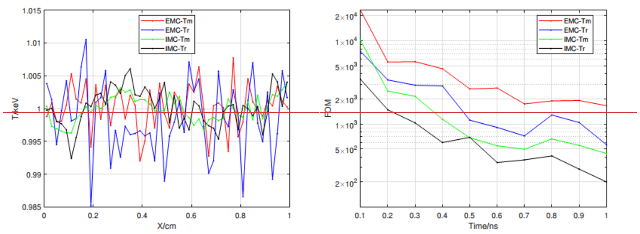
其中，$VAR$ 是方差，$RT$ 是 CPU 运行时间. 更高的 FOM 代表着更好的综合计算

性能. 当模拟时间$t=1.0\,\text{ns}$时，如表 2 所示, IMC 方法耗时 669 秒, EMC 方法仅耗时 53 秒. 由图 4 所示，虽然 EMC 方法的温度变化相较 IMC 方法更加剧烈，但是其 FOM 指标仍表现出明显优势，这表明 EMC 以略高的方差代价换取了计算效率的提升. 在光学厚区域中，IMC 方法引入的伪散射机制会显著延长粒子的有效存活时间，使辐射能量缓慢沉积到物质场中，因而温度更新时的统计波动较小。相较而言, EMC 方法基于宏观方程对发射源项进行估计并直接抽样，大量的粒子权重快速衰减至 0，意味着辐射能量更加迅速地沉积到物质场中，从而使温度更新容易表现出更大的瞬时波动.

表 2　Infinite medium 问题 CPU 运行时间对比.

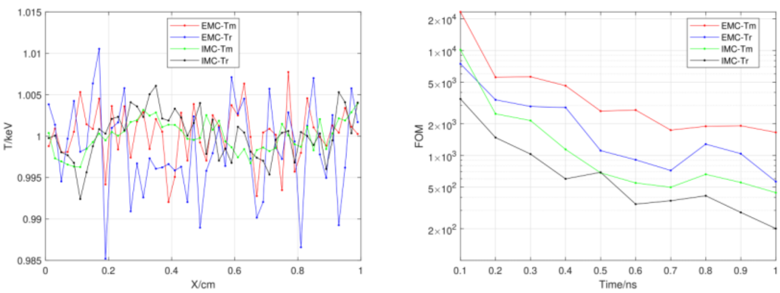
Table 2 CPU runtime comparison of Infinite medium problem.

| 算例 | EMC(秒) | IMC(秒) |
|---|---|---|
| Infinite medium | 53 | 669 |




~~图 4 Infinite medium 问题. 左：温度对比. 右: FOM 指标对比.~~

~~Fig. 4 Infinite medium problem. Left: comparison of temperature. Right: comparison of FOM.~~




(a) (b)

图 4 Infinite medium 问题. (a) 温度对比; (b) FOM 指标对比.

Fig. 4 Infinite medium problem. (a) Comparison of temperature. (b) Comparison of FOM.

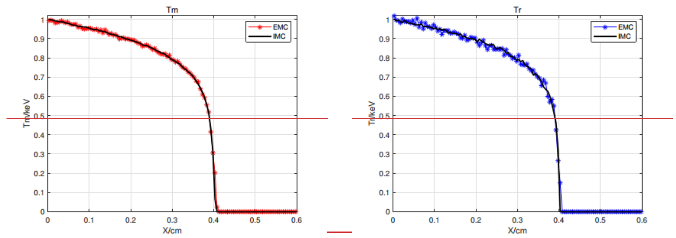
2) Marshark-2B 问题. 本算例求解的是灰体辐射输运方程，数值结果来源于

文献[46]. 由表 3 和图 5 可知, 对于扩散占优的问题, 在参数相同的情况下, EMC 与 IMC 的数值解高度吻合. 尽管 EMC 方法的方差稍大, 但其计算效率具有显著优势.

表 3　Marshark-2B 问题 CPU 运行时间对比.

Table 3 CPU runtime comparison of Mashark-2B problem.

| 算例 | EMC(秒) | IMC(秒) |
| --- | --- | --- |
| Marshark-2B | 1578 | 14694 |




~~图 5 Marshark-2B 问题. 左: 物质温度. 右: 辐射温度.~~

~~Fig. 5 Marshark-2B problem. Left: material temperature. Right: radiation temperature.~~

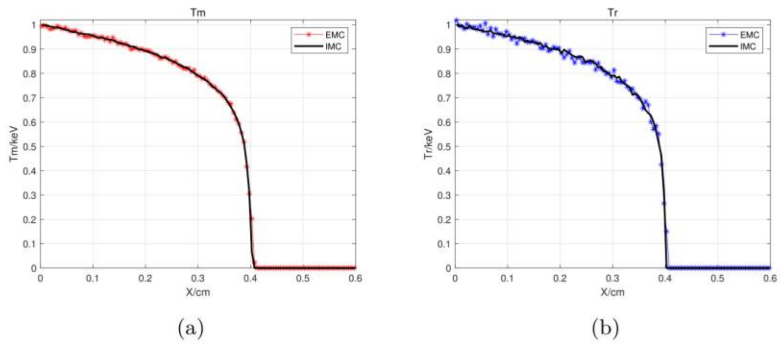



图 5 Marshark-2B 问题. (a) 物质温度; (b) 辐射温度.

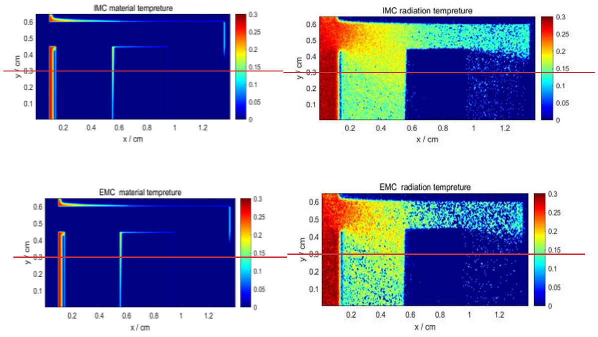
Fig. 5 Marshark-2B problem. (a) Material temperature; (b) Radiation temperature.

3)多群空腔问题. 本算例求解的是频率依赖辐射输运方程, 数值结果来源于文献[47]. 由表 4 和图 6 可知, 对于阻尼系数间断的问题, EMC 与 IMC 在相同参数下的数值解同样高度吻合. 虽然 EMC 的统计波动较大, 但其计算速度显著提升.

表 4 多群空腔问题 CPU 运行时间对比.

Table 4 CPU runtime comparison of the multi-group hohlraum problem.

| 算例 | EMC(秒) | IMC(秒) |
| --- | --- | --- |
| Hohlraum | 17687 | 96332 |




~~图 6 多群空腔问题. 左上: IMC 物质温度. 右上: IMC 辐射温度. 左下: EMC 物质温度. 右下: EMC 辐射温度.~~

~~Fig. 6 Multi-group hohlraum problem. Left upper: IMC material temperature. Right upper: IMC radiation temperature. Left upper: EMC material temperature. Right lower: EMC radiation temperature.~~

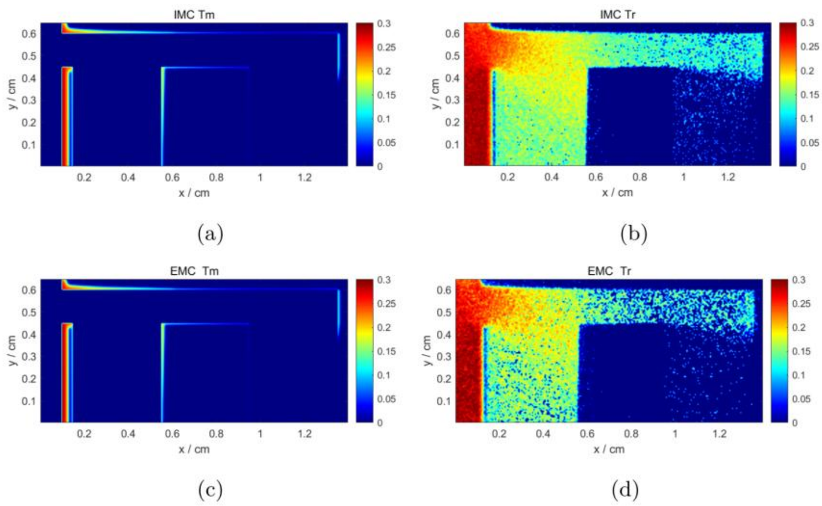



图 6 多群空腔问题. (a) IMC 物质温度; (b) IMC 辐射温度; (c) EMC 物质温度; (d) EMC 辐射温度.

Fig. 6 Multi-group hohlraum problem. (a) IMC material temperature; (b) IMC radiation temperature; (c) EMC material temperature; (d) EMC radiation temperature.

## 3 结论

本文介绍了一类基于特征线回溯思想构造的渐近保持格式, 并展示其在辐射输运问题中的应用. 该方法的核心理念是沿特征线构建尺度自适应的宏观–微

观耦合模型，以还原动理学模型的渐近极限. 在数值计算过程中，先求解宏观方程以获得宏观量的预测值，再求解微观方程并对宏观量进行相容修正，形成闭合的迭代框架. 该方法具备全尺度一致的无条件稳定性，优异的并行效率和良好的扩展性，适合求解高维跨尺度动理学问题.

## 参 考 文 献


[1] POMRANING G C. The ~~Equations~~ equations of ~~Radiation~~ radiation ~~Hydrodynamics~~hydrodynamics[M]. Elsevier Science & Technology, 1973.

[2] BITTENCOURT J A. Fundamentals of ~~Plasma~~ plasma ~~Physics~~physics[M]. New York, NY: Springer, 2004.

[3] MARKOWICH P A, RINGHOFER C A, SCHMEISER C. Semiconductor ~~Equations~~equations[M]. Springer Science & Business Media, 2012.

[4] ALT W. Biased random walk models for chemotaxis and related diffusion approximations[J]. Journal of Mathematical Biology, 1980, 9(2): 147-177.

[5] MARSHAK R E. Note on the Spherical Harmonic ~~Method~~ method ~~As~~ as ~~Applied~~ applied to the Milne ~~Problem~~ problem for a ~~Sphere~~sphere[J]. Physical Review, 1947, 71(7): 443-446.

[6] GRAD H. On the kinetic theory of rarefied gases[J]. Communications on Pure and Applied Mathematics, 1949, 2(4): 331-407.

[7] HAUCK C, MCCLARREN R. Positive $P_N$ ~~Closures~~closures[J]. SIAM Journal on Scientific Computing, 2010, 32(5): 2603-2626.

[8] GARRETT C K, HAUCK C D. A ~~Comparison~~ comparison of ~~Moment~~ moment ~~Closures~~ closures for ~~Linear~~ linear ~~Kinetic~~ kinetic ~~Transport~~ transport ~~Equations~~equations: The ~~Line~~ line ~~Source~~ source ~~Benchmark~~benchmark[J]. Transport Theory and Statistical Physics, 2013, 42(6-7): 203-235.

[9] CAI Z, FAN Y, LI R. Globally Hyperbolic ~~Regularization~~ regularization of Grad’s ~~Moment~~ moment ~~System~~system[J]. Communications on Pure and Applied Mathematics, 2014, 67(3): 464-518.

[10] ALLDREDGE G W, FRANK M, HAUCK C D. A ~~Regularized~~ regularized ~~Entropy~~entropy-~~Based~~ based ~~Moment~~ moment ~~Method~~ method for ~~Kinetic~~ kinetic ~~Equations~~equations[J]. SIAM Journal on Applied Mathematics, 2019, 79(5): 1627-1653.

[11] HUANG J, CHENG Y, CHRISTLIEB A J, et al. Machine learning moment closure models for the radiative transfer equation I: Directly learning a gradient based closure[J]. Journal of Computational Physics, 2022, 453: 110941.

[12] HOU T Y, LI R. Computing nearly singular solutions using pseudo-spectral methods[J]. Journal of Computational Physics, 2007, 226(1): 379-397.

[13] MCCLARREN R G, HAUCK C D. Robust and accurate filtered spherical harmonics expansions for radiative transfer[J]. Journal of Computational Physics, 2010, 229(16): 5597-5614.

[14] CARLSON B G. Solution of the ~~Transport~~ transport ~~Equation~~ equation by Sn ~~Approximations~~approximations: LA-1891[R]. Los Alamos Scientific Lab., N. Mex., 1955.

[15] MILLER W F, REED Wm H. Ray-Effect ~~Mitigation~~ mitigation ~~Methods~~ methods for ~~Two~~two-~~Dimensional~~ dimensional ~~Neutron~~ neutron ~~Transport~~ transport ~~Theory~~theory[J]. Nuclear Science and Engineering, 1977, 62(3): 391-411.

[16] CAMMINADY T, FRANK M, KÜPPER K, et al. Ray effect mitigation for the discrete ordinates method through quadrature rotation[J]. Journal of Computational Physics, 2019, 382: 105-123.

[17] METROPOLIS N, ULAM S. The Monte Carlo ~~Method~~method[J]. Journal of the American Statistical Association, 1949, 44(247): 335-341.

[18] GENTILE N A. Implicit Monte Carlo ~~Diffusion~~diffusion—An ~~Acceleration~~ acceleration ~~Method~~ method for Monte Carlo ~~Time~~tme-~~Dependent~~ dependent ~~Radiative~~ radiative ~~Transfer~~ transfer ~~Simulations~~simulations[J]. Journal of Computational Physics, 2001, 172(2): 543-571.

[19] DENSMORE J D. Interface methods for hybrid Monte Carlo-diffusion radiation-transport simulations[J]. Annals of Nuclear Energy, 2006, 33(4): 343-353

[20] DIMARCO G, PARESCHI L. Hybrid multiscale methods for hyperbolic problems I. Hyperbolic relaxation problems[J]. Communications in Mathematical Sciences, 2006, 4(1): 155-177.

[21] DIMARCO G, PARESCHI L. Hybrid ~~Multiscale~~ multiscale ~~Methods~~ methods II. Kinetic ~~Equations~~equations[J]. Multiscale Modeling & Simulation, 2008, 6(4): 1169-1197.

[22] DIMARCO G, PARESCHI L. Fluid ~~Solver~~ solver ~~Independent~~ independent ~~Hybrid~~ hybrid ~~Methods~~ methods for ~~Multiscale~~ multiscale ~~Kinetic~~ kinetic ~~Equations~~equations[J]. SIAM Journal on Scientific Computing, 2010, 32(2): 603-634

[23] FILBET F, REY T. A ~~Hierarchy~~ hierarchy of ~~Hybrid~~ hybrid ~~Numerical~~ numerical ~~Methods~~ methods for ~~Multiscale~~ multiscale ~~Kinetic~~ kinetic ~~Equations~~equations[J]. SIAM Journal on Scientific Computing, 2015, 37(3): A1218-A1247.

[24] FILBET F, XIONG T. A hybrid discontinuous Galerkin scheme for multi-scale kinetic equations[J]. Journal of Computational Physics, 2018, 372: 841-863.

[25] JIN S. Asymptotic-preserving schemes for multiscale physical problems[J]. Acta Numerica, 2022, 31: 415-489.

[26] JIN S, XIN Z. The relaxation schemes for systems of conservation laws in arbitrary space dimensions[J]. Communications on Pure and Applied Mathematics, 1995, 48(3): 235-276.

[27] BOSCARINO S, PARESCHI L, RUSSO G. Implicit-~~E~~explicit Runge--Kutta ~~S~~schemes for ~~H~~hyperbolic ~~S~~systems and ~~K~~kinetic ~~E~~equations in the ~~D~~diffusion ~~L~~limit[J]. SIAM Journal on Scientific Computing, 2013, 35(1): A22-A51.~~BOSCARINO S. Implicit-Explicit Runge-Kutta Schemes for Hyperbolic Systems in the Diffusion Limit[J]. 2011, 1389: 1315-1318.~~

[28] LEMOU M, MIEUSSENS L. A ~~New~~ new ~~Asymptotic~~ asymptotic ~~Preserving~~ preserving ~~Scheme~~ scheme ~~Based~~ based on ~~Micro~~micro-~~Macro~~ macro ~~Formulation~~ formulation for ~~Linear~~ linear ~~Kinetic~~ kinetic ~~Equations~~ equations in the ~~Diffusion~~ diffusion ~~Limit~~limit[J]. SIAM Journal on Scientific Computing, 2008, 31(1): 334-368.

[29] LEMOU M, MÉHATS F. Micro-~~Macro~~ macro ~~Schemes~~ schemes for ~~Kinetic~~ kinetic ~~Equations~~ equations ~~Including~~ including ~~Boundary~~ boundary ~~Layers~~layers[J]. SIAM Journal on Scientific Computing, 2012, 34(6): B734-B760.

[30] JANG J, LI F, QIU J M, et al. Analysis of ~~Asymptotic~~ asymptotic ~~Preserving~~ preserving DG-IMEX ~~Schemes~~ schemes for ~~Linear~~ linear ~~Kinetic~~ kinetic ~~Transport~~ transport ~~Equations~~ equations in a ~~Diffusive~~ diffusive ~~Scaling~~scaling[J]. SIAM Journal on Numerical Analysis, 2014, 52(4): 2048-2072.

[31] JANG J, LI F, QIU J M, et al. High order asymptotic preserving DG-IMEX schemes for discrete-velocity kinetic equations in a diffusive scaling[J]. Journal of Computational Physics,

2015, 281: 199-224.
[32] PENG Z, CHENG Y, QIU J M, et al. Stability-enhanced AP IMEX-LDG schemes for linear kinetic transport equations under a diffusive scaling[J]. Journal of Computational Physics, 2020, 415: 109485.
[33] PENG Z, CHENG Y, QIU J M, et al. Stability-enhanced AP IMEX1-LDG ~~Method~~method: ~~Energy~~energy-based ~~Stability~~ stability and ~~Rigorous~~ rigorous AP ~~Property~~property[J]. SIAM Journal on Numerical Analysis, 2021, 59(2): 925-954.
[34] PENG Z, LI F. Asymptotic ~~Preserving~~ preserving IMEX-DG-S Schemes for ~~Linear~~ linear ~~Kinetic~~ kinetic ~~Transport~~ transport ~~Equations~~ equations ~~Based~~ based on Schur ~~Complement~~complement[J]. SIAM Journal on Scientific Computing, 2021, 43(2): A1194-A1220.
[35] JIN S, PARESCHI L, TOSCANI G. Uniformly ~~Accurate~~ accurate ~~Diffusive~~ diffusive ~~Relaxation~~ relaxation ~~Schemes~~ schemes for ~~Multiscale~~ multiscale ~~Transport~~ transport ~~Equations~~equations[J]. SIAM Journal on Numerical Analysis, 2000, 38(3): 913-936.
[36] JIN S, PARESCHI L. Asymptotic-Preserving (AP) ~~Schemes~~ schemes for ~~Multiscale~~ multiscale ~~Kinetic~~ kinetic ~~Equations~~equations: a ~~Unified~~ unified ~~Approach~~approach[C]. Basel: Birkhäuser, 2001: 573-582.
[37] FILBET F, JIN S. A class of asymptotic-preserving schemes for kinetic equations and related problems with stiff sources[J]. Journal of Computational Physics, 2010, 229(20): 7625-7648.
[38] JIN S, LI Q. A BGK-penalization-based asymptotic-preserving scheme for the multispecies Boltzmann equation[J]. Numerical Methods for Partial Differential Equations, 2013, 29(3): 1056-1080.
[39] XU K, HUANG J C. A unified gas-kinetic scheme for continuum and rarefied flows[J]. Journal of Computational Physics, 2010, 229(20): 7747-7764.
[40] DIMARCO G, PARESCHI L. Exponential Runge–Kutta ~~Methods~~ methods for ~~Stiff~~ stiff ~~Kinetic~~ kinetic ~~Equations~~equations[J]. SIAM Journal on Numerical Analysis, 2011, 49(5): 2057-2077.
[41] MIEUSSENS L. On the asymptotic preserving property of the unified gas kinetic scheme for the diffusion limit of linear kinetic models[J]. Journal of Computational Physics, 2013, 253: 138-156.
[42] LI Q, PARESCHI L. Exponential Runge–Kutta for the inhomogeneous Boltzmann equations with high order of accuracy[J]. Journal of Computational Physics, 2014, 259: 402-420.
[43] SUN W, JIANG S, XU K. An asymptotic preserving unified gas kinetic scheme for gray radiative transfer equations[J]. Journal of Computational Physics, 2015, 285: 265-279.
[44] ZHANG G, ZHU H, XIONG T. Asymptotic ~~Preserving~~ preserving and ~~Uniformly~~ uniformly ~~Unconditionally~~ unconditionally ~~Stable~~ stable ~~Finite~~ finite ~~Difference~~ difference ~~Schemes~~ schemes for ~~Kinetic~~ kinetic ~~Transport~~ transport ~~Equations~~equations[J]. SIAM Journal on Scientific Computing, 2023, 45(5): B697-B730.
[45] CAI Y, ZHANG G, ZHU H, et al. Asymptotic preserving semi-Lagrangian discontinuous Galerkin methods for multiscale kinetic transport equations[J]. Journal of Computational Physics, 2024, 513: 113190.
[46] SHI Y, SONG P, XIONG T. An efficient asymptotic preserving Monte Carlo method for radiative transfer equations[J]. Journal of Computational Physics, 2023, 493: 112483.
[47] HONG Y, SHI Y, CAI Y, et al. An efficient asymptotic preserving Monte Carlo method for

frequency-dependent radiative transfer equations[DB/OL]. arXiv, 2025. http://arxiv.org/abs/2507.02392.

[48] FENG Z, XIONG T, TANG M. The decomposed HOLO scheme: Enabling large time steps for the gray radiative transfer equation across three distinct limits[J]. Journal of Computational Physics, 2025, 537: 114092.

[49] ROBERT A. A stable numerical integration scheme for the primitive meteorological equations[J]. Atmosphere-Ocean, 1981, 19(1): 35-46.

[50] STANIFORTH A, CÔTÉ J. Semi-Lagrangian ~~Integration~~ integration ~~Schemes~~ schemes for ~~Atmospheric~~ atmospheric ~~Models~~models—A Review[J]. 1991, 119(9): 2206-2223.

[51] KOPP H J. Synthetic ~~Method~~ method ~~Solution~~ solution of the ~~Transport~~ transport ~~Equation~~equation*[J]. Nuclear Science and Engineering, 1963, 17(1): 65-74.

[52] GOL'DIN V Ya. A quasi-diffusion method of solving the kinetic equation[J]. USSR Computational Mathematics and Mathematical Physics, 1964, 4(6): 136-149.

[53] ALCOUFFE R E. Diffusion ~~Synthetic~~ synthetic ~~Acceleration~~ acceleration ~~Methods~~ methods for the ~~Diamond~~diamond-~~Differenced~~ differenced ~~Discrete~~discrete-~~Ordinates~~ ordinates ~~Equations~~equations[J]. Nuclear Science and Engineering, 1977, 64(2): 344-355.

[54] LORENCE JR. L J, MOREL J E, LARSEN E W. An S2 Synthetic ~~Acceleration~~ acceleration ~~Scheme~~ scheme for the ~~One~~one-~~Dimensional~~ dimensional Sn ~~Equations~~ equations with ~~Linear~~ linear ~~Discontinuous~~ discontinuous ~~Spatial~~ spatial ~~Differencing~~differencing[J]. Nuclear Science and Engineering, 1989, 101(4): 341-351.

[55] GUTHRIE B, HOLLOWAY J P, PATTON B W. GMRES as a multi-step transport sweep accelerator[J]. Transport Theory and Statistical Physics, 1999, 28(1): 83-102.

[56] PATTON B W, HOLLOWAY J P. Application of preconditioned GMRES to the numerical solution of the neutron transport equation[J]. Annals of Nuclear Energy, 2002, 29(2): 109-136.

[57] MANTEUFFEL T, MCCORMICK S, MOREL J, et al. A ~~Fast~~ fast ~~Multigrid~~ multigrid ~~Algorithm~~ algorithm for ~~Isotropic~~ isotropic ~~Transport~~ transport ~~Problems~~ problems I: Pure ~~Scattering~~scattering[J]. SIAM Journal on Scientific Computing, 1995, 16(3): 601-635.

[58] MANTEUFFEL T, MCCORMICK S, MOREL J, et al. A ~~Fast~~ fast ~~Multigrid~~ multigrid ~~Algorithm~~ algorithm for ~~Isotropic~~ isotropic ~~Transport~~ transport ~~Problems~~problems. II: With ~~Absorption~~absorption[J]. SIAM Journal on Scientific Computing, 1996, 17(6): 1449-1474.

[59] ADAMS M L, LARSEN E W. Fast iterative methods for discrete-ordinates particle transport calculations[J]. Progress in Nuclear Energy, 2002, 40(1): 3-159.

[60] EINKEMMER L. Semi-Lagrangian Vlasov simulation on GPUs[J]. Computer Physics Communications, 2020, 254: 107351.

[61] EINKEMMER L, MORIGGL A. A ~~Semi~~semi-Lagrangian ~~Discontinuous~~ discontinuous Galerkin ~~Method~~ method for ~~Drift~~drift-~~Kinetic~~ kinetic ~~Simulations~~ simulations on GPUs[J]. SIAM Journal on Scientific Computing, 2024, 46(2): B33-B55.

[62] QIU J M, SHU C W. Conservative high order semi-Lagrangian finite difference WENO methods for advection in incompressible flow[J]. Journal of Computational Physics, 2011, 230(4): 863-889.

[63] CAI X, GUO W, QIU J M. Comparison of ~~Semi~~semi-Lagrangian ~~Discontinuous~~ discontinuous Galerkin ~~Schemes~~ schemes for ~~Linear~~ linear and ~~Nonlinear~~ nonlinear ~~Transport~~ transport ~~Simulations~~simulations[J]. Communications on Applied Mathematics and

Computation, 2022, 4(1): 3-33.

[64] ZHENG N, CAI X, QIU J M, et al. A fourth-order conservative semi-Lagrangian finite volume WENO scheme without operator splitting for kinetic and fluid simulations[J]. Computer Methods in Applied Mechanics and Engineering, 2022, 395: 114973.

[65] WOLLABER A B. Four ~~Decades~~ decades of ~~Implicit~~ implicit Monte Carlo[J]. Journal of Computational and Theoretical Transport, 2016, 45(1-2): 1-70.

[66] FLECK J A, CUMMINGS J D. An ~~Implicit~~ implicit Monte Carlo ~~Scheme~~ scheme for ~~Calculating~~ calculating ~~Time~~ time and ~~Frequency~~ frequency ~~Dependent~~ dependent ~~Nonlinear~~ nonlinear ~~Radiation~~ radiation ~~Transport~~transport[J]. Journal of Computational Physics, 1971, 8(3): 313-342.

[67] DING M, QIU J M, SHU R. Accuracy and ~~Stability~~ stability ~~Analysis~~ analysis of the ~~Semi~~semi-Lagrangian ~~Method~~ method for ~~Stiff~~ stiff ~~Hyperbolic~~ hyperbolic ~~Relaxation~~ relaxation ~~Systems~~ systems and ~~Kinetic~~ kinetic BGK ~~Model~~model[J]. Multiscale Modeling & Simulation, 2023, 21(1): 143-167.

[68] DING M, QIU J M, SHU R. Semi-Lagrangian nodal discontinuous Galerkin method for the BGK model[J]. Advances in Computational Science and Engineering, 2024, 2(3): 222-245.

[69] CAI X, HAO Z, LIU L, et al. Asymptotic-preserving semi-Lagrangian discontinuous Galerkin schemes for the Boltzmann equation[DB/OL]. arXiv, 2025. http: //arxiv.org/abs/2510.14375.

[70] LIU H, CAI X, CAO Y, et al. An asymptotic-preserving conservative semi-Lagrangian scheme for the Vlasov-Maxwell system in the quasi-neutral limit[J]. Journal of Computational Physics, 2025, 528: 113840.

[71] GUO W, NAIR R D, QIU J M. A ~~Conservative~~ conservative ~~Semi~~semi-Lagrangian ~~Discontinuous~~ discontinuous Galerkin ~~Scheme~~ scheme on the ~~Cubed~~ cubed ~~Sphere~~sphere[J]. Monthly Weather Review, 2014, 142(1): 457-475.

[72] CAI X, GUO W, QIU J M. A ~~High~~ high ~~Order~~ order ~~Conservative~~ conservative ~~Semi~~semi-Lagrangian ~~Discontinuous~~ discontinuous Galerkin ~~Method~~ method for ~~Two~~two-~~Dimensional~~ dimensional ~~Transport~~ transport ~~Simulations~~simulations[J]. Journal of Scientific Computing, 2017, 73(2): 514-542.

[73] CAI X, BOSCARINO S, QIU J M. High order semi-Lagrangian discontinuous Galerkin method coupled with Runge-Kutta exponential integrators for nonlinear Vlasov dynamics[J]. Journal of Computational Physics, 2021, 427: 110036.

[74] SEIBOLD B, FRANK M. StaRMAP---A ~~Second~~ second ~~Order~~ order ~~Staggered~~ staggered ~~Grid~~ grid ~~Method~~ method for ~~Spherical~~ spherical ~~Harmonics~~ harmonics ~~Moment~~ moment ~~Equations~~ equations of ~~Radiative~~ radiative ~~Transfer~~transfer[J]. ACM Transactions on Mathematical Software, 2014, 41(1): 1-28.

# Efficient Asymptotic-Preserving Method Based on Characteristics for Kinetic Equations

CAI Yi[1], HONG Yiyang[1], XIONG Tao[2,*]

*(1. School of Mathematical Sciences, Xiamen University, Xiamen, Fujian 361005, ~~PR~~ China;*

*2. School of Mathematical Sciences, University of Science and Technology of China, Hefei, Anhui 230026, ~~PR~~ China*)

**Abstract:** Kinetic equations serve as fundamental mathematical models for describing the dynamic behavior of particle systems. Their intrinsic multiscale structure, however, poses significant challenges in developing efficient and robust numerical schemes. A popular approach is to design asymptotic preserving (AP) schemes. It aims to create a unified solver across the computational domain that captures the correct asymptotic limits at a discrete level. In recent years, we have developed a class of unconditionally stable AP schemes based on characteristic backtracking. These methods feature low computational complexity, high parallel efficiency, and excellent scalability, making them particularly suitable for long-time simulations of high-dimensional, multiscale transport phenomena. Owing to these advantages, the proposed framework holds great potential for applications in national defense engineering, astrophysics, semiconductor, and biochemistry.

**Keywords:** kinetic equation; parallel computing; characteristic backtracking; asymptotic preserving